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Maternal and lactogenic immunity in gestating sows induced by adenoviruses expressing mucosal immunogens derived from porcine epidemic diarrhea virus

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Abstract

Porcine epidemic diarrhea virus (PEDV) is a significant pathogen that causes severe diarrhea and high mortality rates in piglets. Thus, maternal and lactogenic immunity is a key success in protecting piglets from PEDV. Here, we developed four recombinant adenovirus (rAd)-based vaccine candidates against PEDV harboring novel immunogens fused with mucosal adjuvants and evaluated their capacity to elicit maternal and lactogenic immunity in gestating sows. The rAd-based vaccines were developed on the basis of the new immunogen PEDVSME (rAd. PEDVSME) and its derivatives fused with three mucosal adjuvants: bacterial outer membrane protein H (OmpH), cholera toxin B subunit (CTB), and GM-CSF/IL-4 fusion protein (GI). In a randomized controlled trial, a total of 50 pregnant sows (n = 10/group) received a prime-boost vaccination regimen of rAd. PEDVSME, rAd. PEDVSME-OmpH, rAd. PEDVSME-CTB, rAd. PEDVSME-GI and PBS were used as controls. After the second dose, the rAd. PEDVSME-CTB induced the highest PEDV-specific IgG response with the highest PEDV-neutralizing titer in pregnant sows, whereas rAd. PEDVSME-OmpH elicited the greatest level of systemic PEDV-specific IgA responses. For the transfer of maternal PEDV-specific antibodies into colostrum, all three rAd-based vaccines expressing adjuvanted immunogens (PEDVSME-OmpH, PEDVSME-CTB, PEDVSME-GI) were superior to the rAd expressing the original immunogen PEDVSME and the PBS control. Interestingly, IgG was the dominant isotype in colostrum, and correlated more strongly with neutralizing activity than IgA. In offspring, newborn piglets from all four groups of sows receiving rAd-based vaccines had antibodies with neutralizing titers higher than those from the control group. During the weaning period, decreases in neutralizing titers were observed in all groups, except for piglets from the rAd group. PEDVSME-OmpH group, whose neutralizing titers were well maintained and significantly greater than those in the control group ($P<0.05$). These findings demonstrate that the rAd-based vaccines expressing the PEDVSME immunogen fused with the mucosal adjuvant OmpH (rAd.

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PEDVSME-OmpH) are primary candidates for further evaluation viral challenge in piglets to determine their protective efficacy via passively lactogenic immunity.

Keywords Lactogenic immunity, Mucosal adjuvant, Neutralizing antibody, PEDV, Recombinant adenovirus vaccine

Introduction

Porcine epidemic diarrhea (PED) is an infectious disease with the etiology of porcine epidemic diarrhea virus (PEDV), which has burdened and negatively impacted the swine industry worldwide (Jung et al. 2020; Zhang et al. 2022). PEDV is categorized in the subgenus *Pedecovirus*, genus *Alphacoronavirus*, family *Coronaviridae*, and order *Nidovirales* (Hanke et al. 2015; Lin et al. 2016). Although PEDV can infect pigs of all ages, it causes severe diarrhea in piglets, particularly newborns, resulting in a mortality rate as high as 80–100% (Madson et al. 2014; Fu et al. 2017). PEDV can be classified into two genogroups on the basis of the genetic diversity of the spike (S) gene: G1 (classical) and G2 (variant) (Fu et al. 2017; Jang et al. 2023; Su et al. 2023b). The G1 genotype includes two subtypes, G1a and G1b, whereas the G2 genotype includes three subtypes, G2a, G2b, and G2c (Guo et al. 2019). In Thailand, PED first emerged in 2007, with outbreaks caused by a genogroup 2 variant (Temeeyasen et al. 2014; Stott et al. 2017; Jermsutjarit et al. 2024). Currently, PED has become endemic, with many herds experiencing recurrent outbreaks involving both PEDV genogroups 1 and 2 (Cheun-Arom et al. 2015; Stott et al. 2017). In 2014–2015, several herds reported severe diarrhea outbreaks (Janetanakit et al. 2016; Stott et al. 2017).

The RNA genome is approximately 28 kb in length and contains 5' ends, 5' end caps, 5'-untranslated regions (5'-UTRs), 7 open reading frames (ORFs), 3'-untranslated regions (3'-UTRs), and 3'-polyadenylated (polyA) tail structures at the 3' end (Kocherhans et al. 2001; Lin et al. 2022; Liao et al. 2023; Jang et al. 2023). ORF1a and ORF1b encode nonstructural proteins for replication and transcription, and ORFs 2, 4, 6, and 7 encode the structural proteins spike (S), envelope (E), membrane (M), and nucleocapsid (N), respectively (Kocherhans et al. 2001; Song and Park 2012; Feng et al. 2023). The S glycoprotein plays a crucial role in viral entry, cell adaptation, and virulence (Wang et al. 2020; Li et al. 2020). It is composed of S1 and S2 subunits, by which S1 is responsible for interacting with host cellular receptors to mediate viral entry, and S2 facilitates the fusion of the virus with host cells (Wang et al. 2020; Li et al. 2020). Thus, the PEDV-S protein is considered a primary immunogen and ideal target for alternative vaccine development against PEDV (Su et al. 2023a; Murtaza et al. 2024; Zhao et al. 2024). The S1 subunit consists of a neutralizing epitope region distinct from the collagenase equivalent domain (COE) located at residues 499–638 (Murtaza et al. 2024). As the S1 subunit is a large and complex protein, COE is a preferred antigen for PEDV vaccine development (Ma et al. 2018; Li et al. 2023). A COE-based vaccine efficiently induced neutralizing antibodies against PEDV (Chang et al. 2002). Moreover, several neutralizing epitopes have also been identified, including epitopes M (Cao et al. 2015), SS2, SS6 (Sun et al. 2008), and 2C10 (Cruz et al. 2006). Recently, more B-cell epitopes across the S and M proteins have been identified, together with potential neutralizing epitopes (Polyiam et al. 2022).

PEDV infects and replicates mainly in enterocytes in the intestine; thus, PED is recognized as a mucosal disease (Lee 2015). Upon PEDV infection, the level of PEDV-specific immunoglobulin G and A (IgG and IgA) antibodies is significantly increased in the serum and intestine (Krishna et al. 2020). The relevant literature summarizes the crucial role of mucosal immunity in inhibiting enteric viral replication. Additionally, the recovery period or postinfection period in pregnant sows involves the migration of gut-derived IgA immunocytes (B memory cells) to terminate at the mammary glands. Consequently, high secretory IgA (sIgA) antibody (Ab) titers have been noted in colostrum and milk (gut-mammary-sIgA axis) (Salmon et al. 2009; Langel et al. 2016). This phenomenon contributes the greatest benefit to neonatal piglets from passive lactogenic immunity (Lin et al. 2019), thus emphasizing the importance of PEDV-specific sIgA and IgG in pregnant sows. For effective protection of piglets against PEDV, a vaccine that can stimulate PEDV-specific intestinal mucosal immunity is highly desirable (Lee 2015).

However, promising vaccine candidates are controversial because of their inability to induce lactogenic immunity and cross-protection. Recently, several studies have revealed the development of PEDV vaccine candidates via various platforms. Inactivated (killed) vaccines and live-attenuated vaccines are incapable of providing complete protection for the study groups during a vaccination-challenge trial with highly pathogenic PEDV (Lin et al. 2019; Krishna et al. 2020; Song et al. 2024). Therefore, more effective vaccines with other platforms are still needed. The adenovirus vector system is a versatile and regularly utilized vector for novel vaccine development and gene therapy (Do et al. 2020). Recombinant adenovirus (rAd) vectors offer several advantages over whole-virus vaccines. These advantages include strong

immunogenicity, a large capacity for foreign DNA, and the ability to infect many cell types (Kamen and Henry 2004; Geisbert et al. 2011). The rAd vaccines can be produced at high titers, are stable for long-term storage, do not integrate into the host genome, and rapidly express transgenes for immune responses (Vorburger and Hunt 2002; Gebre et al. 2021). Additionally, the well-understood biology of adenoviruses has facilitated faster development and regulatory approval. Human adenovirus type 5 (Ad5) is the most widely used adenoviral vector because of its established nature and accessibility. The Ad5 vector is a replication-defective virus capable of eliciting robust systemic and mucosal immune responses (Liu et al. 2019; Senthilkumaran et al. 2022; Wang et al. 2023). Previous studies have also demonstrated that the Ad5 vector vaccine can be expressed in pigs (Senthilkumaran et al. 2022), and it has been used in the development of vaccines against pig pathogens such as African swine fever virus (ASFV) (Lokhandwala et al. 2019; Cadenas-Fernández et al. 2020), PEDV (Liu et al. 2019; Do et al. 2020), porcine reproductive and respiratory syndrome virus (PRRSV) (Xie et al. 2019), and porcine circovirus (PCV) (Zhang et al. 2019). This approach was also our choice for PEDV vaccine delivery in this study.

With the goal of achieving a more potent PEDV vaccine, we previously designed a new immunogen, termed PEDVSME. This immunogen contains the COE domain fused with four reported neutralizing epitopes from the S protein: M, SS2, SS6, and 2C10. To further increase the capacity to stimulate mucosal immunity against PEDV, we conjugated PEDVSME with mucosal adjuvants, including (i) bacterial outer membrane protein H (OmpH), (ii) cholera toxin B subunit (CTB), and (iii) granulocyte–macrophage colony-stimulating factor (GM-CSF) and interleukin-4 (IL-4) fusion protein (GI). The synthetic genes encoding these four immunogens were incorporated into the Ad5 vector, and recombinant Ad5 viruses were generated and amplified in HEK293A cells. Furthermore, the immunogenicity of all four rAd-based PEDV vaccines was evaluated in sows and piglets, with a focus on assessing the transfer of maternal antibodies to piglets. Antibodies (both IgG and IgA), along with their neutralizing activity, were measured in sow sera, colostrum, milk, and piglet sera. In addition, we investigated the cellular immune response via ELISpot. Finally, we analyzed the correlation of the antibody response with neutralizing activity.

Results

Immunogen design and generation of rAd-based vaccines

In this work, a new immunogen, namely, PEDVSME and its variants, which we previously designed, used in the generation of a recombinant adenovirus. This immunogen is a chimeric protein composed of the core neutralizing epitope region COE (aa 488–645) and four neutralizing B-cell epitopes, M, SS2, SS6, and 2C10, from the PEDV S protein (Fig. 1A). With an emphasis on the induction of mucosal immunity, the immunogen has been modified with mucosal adjuvants, OmpH, CTB, and the GM-CSF/IL4 (GI) fusion. These give rise to three immunogen variants: PEDVSME-OmpH, PEDVSME-CTB, and PEDVSME-GI. All the constructs are tagged with a C-terminal V5 epitope to facilitate protein detection and analysis. The synthetic genes of all four constructs were subjected to rescue recombinant adenovirus, prior to virus amplification and purification as described in Fig. 1B. Following transfection into HEK293 cells, we observed adenovirus-induced cytopathic effects, confirming successful viral production.

Protein expression by rAd

The expression of the immunogen by rAd was validated through immunofluorescence and Western blot analyses. Expression was studied in HEK293A and PK-15 cells to assess the versatility of the adenoviral vector system. Following infection with rAd at an MOI of 10, HEK293A cells robustly expressed all four immunogen constructs, as evidenced by positive staining with both anti-V5 (red) and anti-adenovirus (green) antibodies, but not in uninfected cells (Fig. 2A). To evaluate the ability of rAd in a more relevant cellular context, we examined immunogen expression in PK-15 cells infected at an MOI of 100. Strong immunofluorescent signals were detected using an anti-V5 antibody (red) for all four constructs, indicating efficient protein expression in porcine cells (Fig. 2B). The expression of the GI fusion protein was investigated by costaining the rAd. PEDVSME-GI-infected PK-15 cells with anti-V5 (red) and anti-IL-4 (green) antibodies presented the presence of the GI fusion protein (green) within the cells expressing PEDVSME (red), confirming the successful coexpression of these two proteins by rAd. PEDVSME-GI in pig cells (Fig. 2C). Western blot analysis provided further confirmation of immunogen expression, revealing distinct protein bands at the expected molecular weights for all four constructs in PK-15 cells infected with rAd at an MOI of 100 (Fig. 2D). The clear, single bands observed for each construct suggest proper protein processing and stability in mammalian cells. However, for the fusion protein, GI could not be detected by Western blot.

Immunization of gestating sows

To study the immunogenicity of adenoviral vaccine candidates, a randomized controlled study was conducted with 50 gestating sows (n = 10/group). Four experimental groups were immunized via intramuscular injection

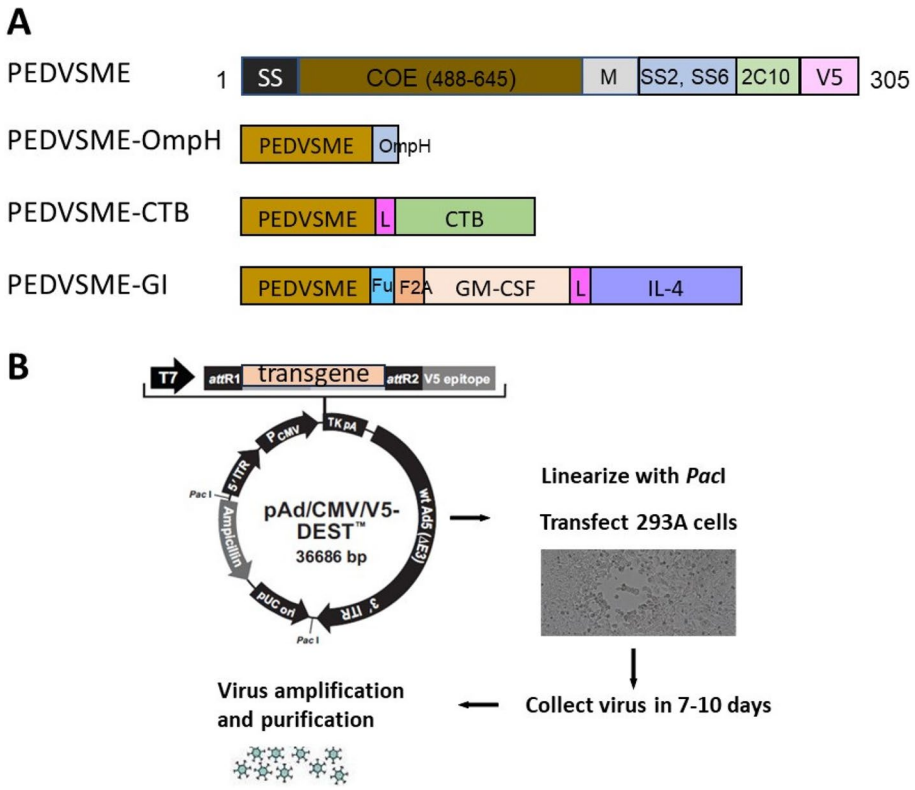

Fig. 1 Schematic diagram showing the designed immunogens and generation of rAd. **A** Composition of the PEDVSME immunogen and its derivatives: PEDVSME-OmpH, PEDVSME-CTB, and PEDVSME-GI. L: linker; Fu: furin site; F2A: self-cleaving peptide F2A. **B** Schematic diagram representing the generation of rAd

with different rAd-based vaccines: rAd. PEDVSME (Group 1; G1), rAd. PEDVSME-OmpH (Group 2; G2), rAd. PEDVSME-CTB (Group 3; G3), rAd. PEDVSME-GI (Group 4; G4) and PBS control group (Group 5; G5) (Fig. 3A). The immunization schedule followed a prime–boost regimen, with the primary dose (prime) administered at 10 weeks of gestation, followed by a booster dose at 13 weeks of gestation. All sows maintained normal gestation and farrow at approximately week 16. To evaluate both maternal immunity and passive immunity, we collected sow serum, colostrum/milk, and piglet serum at different time points. Sow serum samples were collected at three time points: prevaccination (week 10), postprime (week 13), and postboost (week 16). Colostrum/milk samples were collected at the time of farrowing (designated 'colostrum') and 5–9 days postfarrowing (designated 'milk') to evaluate lactogenic immunity. To assess passive immunity in offspring, piglet serum samples were collected at two time points: after birth (week 16) and at weaning (week 20). The complete experimental timeline and sampling approach are detailed in Fig. 3B.

Serological antibody responses in immunized sows

IgG and IgA responses in sow sera at preimmunization and after the second dose were measured via a commercial ELISA kit and in-house ELISA tests based on the recombinant COE and PEDVSME antigens. Analysis of covariance (ANCOVA) was performed to compare postimmunization antibody responses among the groups. Preimmunization levels were used as a covariate in the model to statistically control for baseline differences, thereby ensuring a more precise comparison of the vaccine-induced responses. The adjusted mean serum IgG levels for all experimental groups are presented in Fig. 4A. When assessed with the commercial ELISA kit and the COE-based in-house ELISA, no statistically significant differences in IgG levels were detected among any of the immunized groups (G1–G4) and the control group (G5). In contrast, analysis via the PEDVSME-based ELISA revealed significant differences in the magnitude of the IgG response. Notably, rAd. PEDVSME-CTB (G3) elicited the most robust IgG response, which was significantly greater than that in the G1, G2, and control. Furthermore, the rAd. PEDVSME-GI (G4) also showed significantly elevated IgG when compared to the G2 and the control.

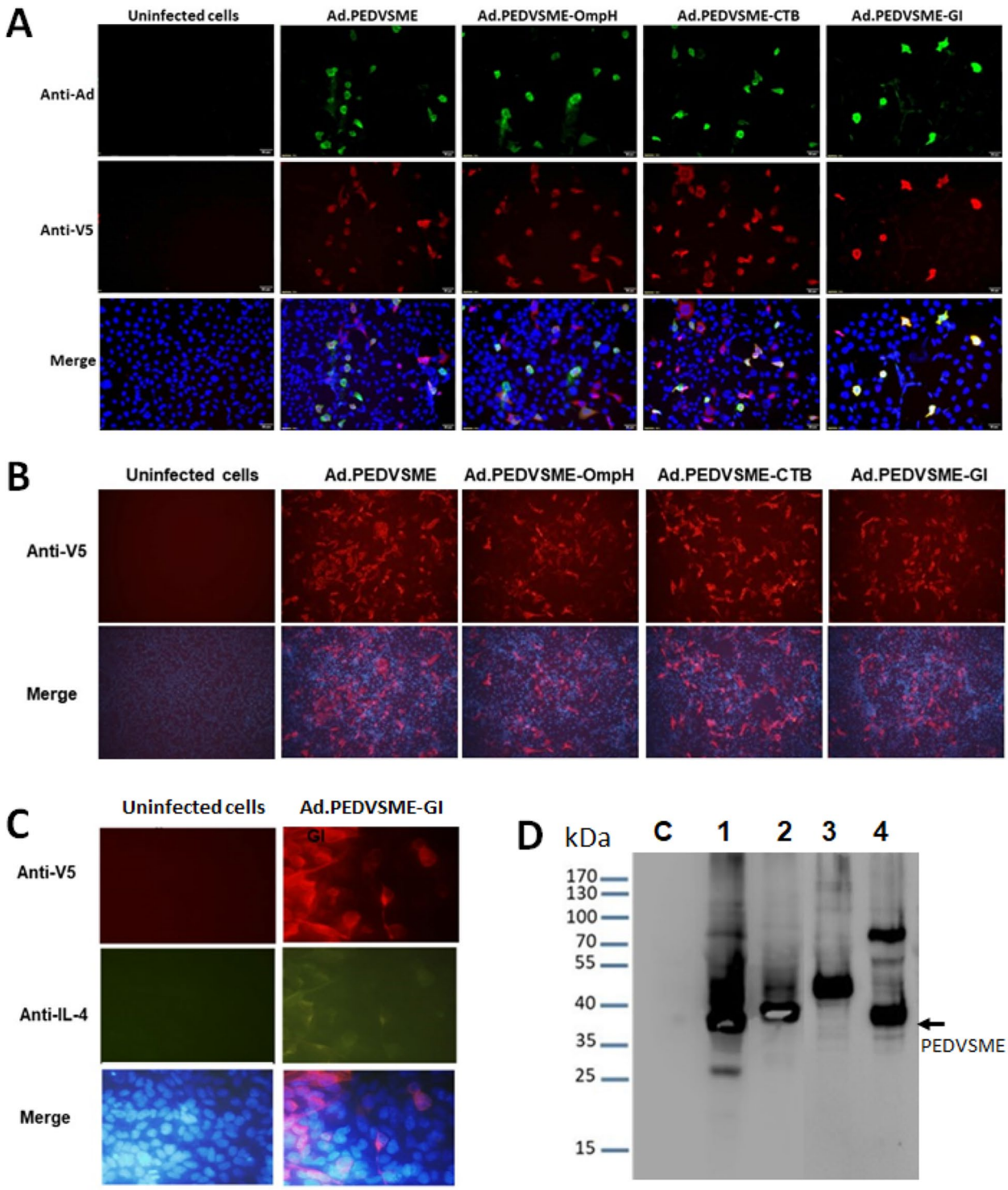

Fig. 2 Verification of PEDVSME and derivative protein expression by recombinant adenovirus vectors in mammalian cells. Immunofluorescence detection utilized an anti-V5 antibody (red) for PEDVSME immunogen visualization, an anti-adenovirus antibody (green) for viral protein detection, and an anti-IL-4 antibody (green) for GI fusion protein detection. The number of nuclei was determined via DAPI staining (blue fluorescence). **A** Immunofluorescence analysis of HEK293A cells infected with rAds at an MOI of 10. **B** Immunofluorescence visualization of rAd-transduced PK-15 cells at an MOI of 100. **C** Subcellular colocalization analysis of the PEDVSME immunogen and GI in PK-15 cells transduced with rAd. PEDVSME-GI. **D** Western blot characterization of immunogen expression in PK-15 cells. Lanes 1–4 represent cells transduced with rAds expressing PEDVSME, PEDVSME-OmpH, PEDVSME-CTB, or PEDVSME-GI, respectively. Lane C: untransduced control cells. The arrow indicates the expected molecular weight of the PEDVSME protein

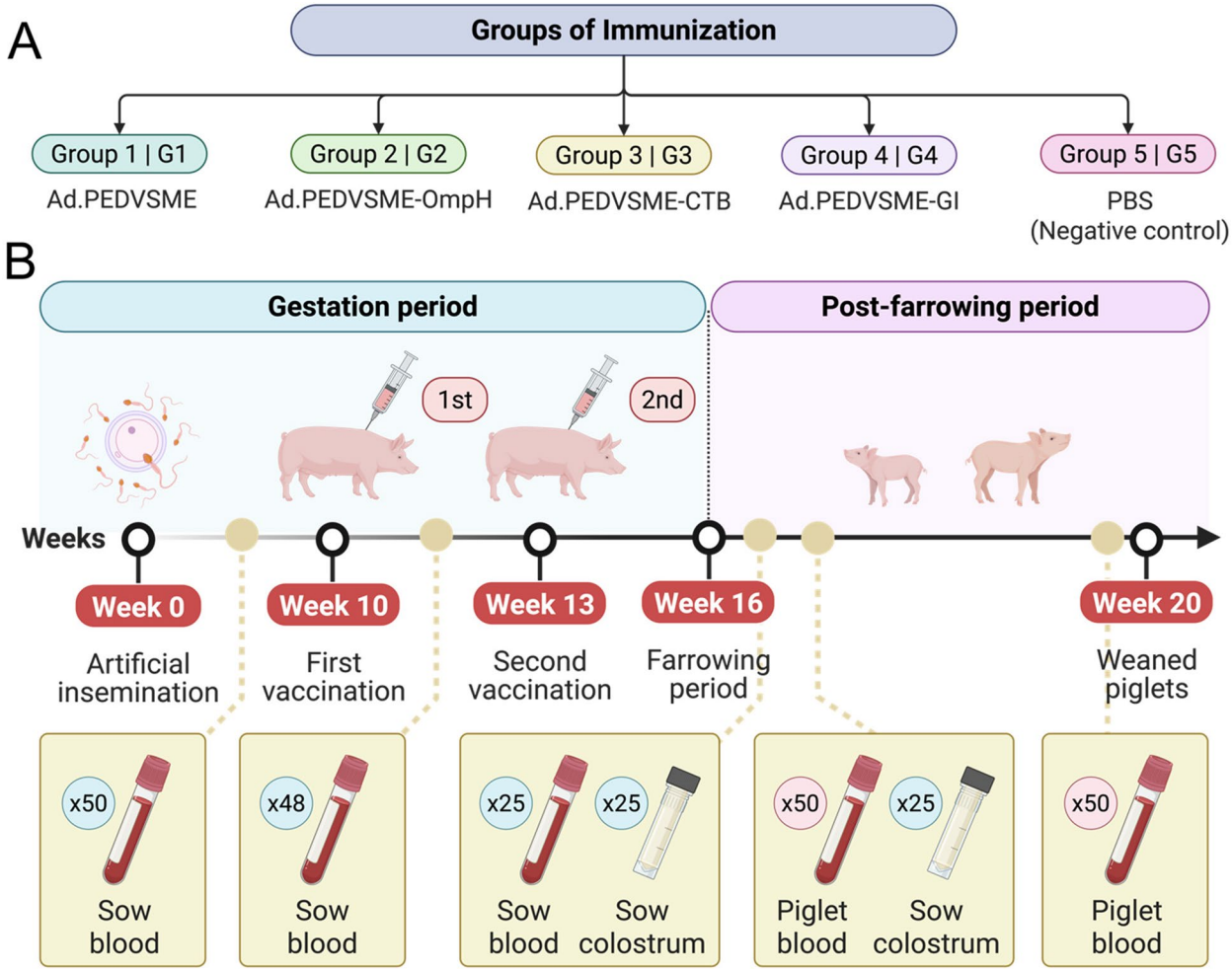

Fig. 3 Immunization protocol. **A** Immunization groups. **B** Experimental timeline and sampling scheme for the evaluation of PEDV vaccine efficacy in pregnant sows. The figure was created using BioRender.com

For IgA responses, adjusted mean serum IgA levels for all experimental groups are illustrated in Fig. 4B. The overall IgA responses were generally moderate across all groups and ELISA tests. By using commercial and PEDVSME-based ELISA tests, the rAd. PEDVSME-OmpH (G2) presented the highest IgA level. When measured with the COE-based ELISA, there were no statistically significant differences in IgA levels among the experimental groups. In the PEDVSME-based ELISA, the rAd. PEDVSME-OmpH (G2) and with the rAd. PEDVSME-GI (G4) group ($P < 0.05$). No other statistically significant differences in IgA responses were detected among the groups.

Taken together, among the rAd-based vaccines expressing adjuvanted PEDVSME vaccines, rAd. PEDVSME-CTB (G3) was the most effective at inducing a systemic PEDV-specific IgG response, whereas rAd. PEDVSME-OmpH (G2) was superior to other vaccines in eliciting systemic IgA responses in gestating sows.

Lactogenic immunity passively acquired from immunized sows

Antibody responses in mammary secretions were investigated in colostrum (collected during the farrowing period) and milk (collected 5–9 days postfarrowing). Three ELISA tests revealed distinct patterns among the different groups and sampling time points. The commercial ELISA kit (Innoceleris) showed only low IgG responses in both the colostrum and milk samples, with no difference between the immunized groups (G1–G4) and the control group (G5) (Fig. 5A, left panel). However, when COE- and PEDVSME-based ELISA tests were used, although none of the sample groups showed a statistically significant difference in IgG levels compared with the samples from the control group (G5), there was

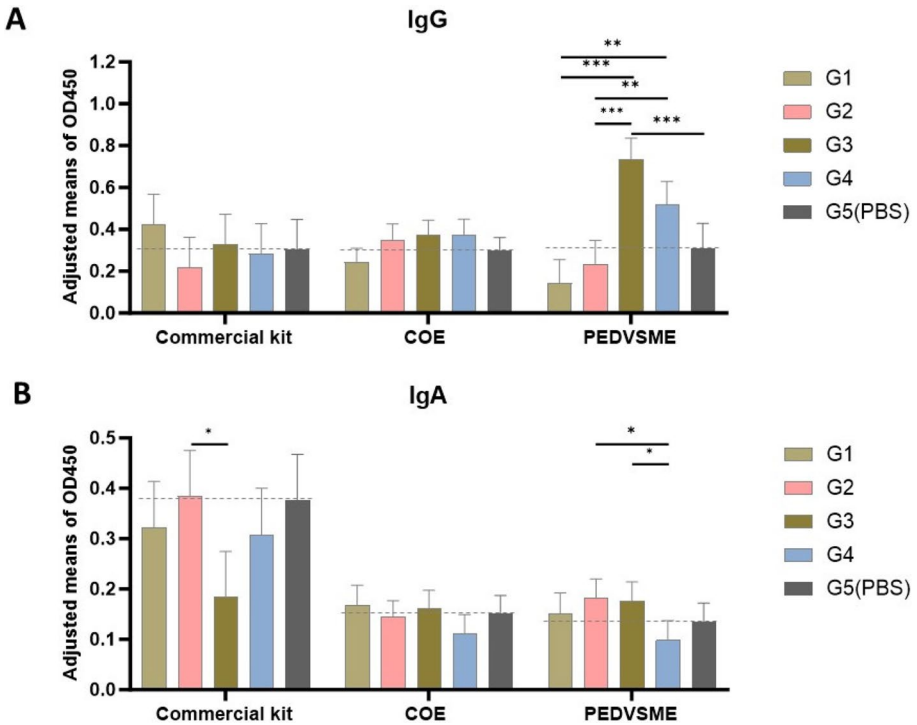

Fig. 4 Serum IgG and IgA responses in gestating sows following immunization. Gestating sows were immunized with four different adenovirus vaccines: rAd. PEDVSME (G1), rAd. PEDVSME-OmpH (G2), rAd. PEDVSME-CTB (G3), and rAd. PEDVSME-GI (G4). The control group (G5). **A** Adjusted mean IgG levels. **B** Adjusted mean IgA levels. Bars represent adjusted means ± standard errors of the means (SEMs). Asterisks denote statistically significant differences between the immunized groups (G1–G4) and the control group (G5) following ANCOVA with post hoc comparisons: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

a trend toward greater responses in some immunized groups. At farrowing, colostrum from all four immunized groups showed robust IgG responses (OD450 values ranging from 0.4–0.5), which were greater than those in the control group (G5), with the highest IgG levels in the rAd. PEDVSME-GI (G4) group. Postfarrowing milk samples presented a general decline in IgG levels in all groups. (Fig. 5A, right panel).

Regarding IgA responses investigated by commercial kits (Innoceleris and IDEXX), most groups, including the control group (G5), presented high levels of IgA in milk, both farrowing and postfarrowing (Fig. 5B). However, only the Innoceleris ELISA kit demonstrated greater IgA responses in milk samples from sows immunized with rAd. PEDVSME (G1), rAd. PEDVSME-OmpH (G2) and rAd. PEDVSME-CTB (G3) than did the control group (G5) at the farrowing period (Fig. 5B, left panel). COE-based ELISA revealed a similar trend in the IgA response obtained from the Innoceleris ELISA kit in colostrum samples, although the OD450 values were generally lower. Interestingly, during the postfarrowing period, IgA levels in milk samples from all immunized groups (G1–G4) were lower than those in milk samples from the control group (G5) (Fig. 5B, right panel). Taken together, these results suggest that rAd. PEDVSME (G1), rAd. PEDVSME-OmpH (G2) and rAd. PEDVSME-CTB (G3) efficiently induced the secretion of both IgG and IgA in milk, whereas rAd. PEDVSME-GI (G4) elicited IgA in milk.

Passive immunity in piglets

To comprehensively evaluate passive immunity transfer from the immunized sows to their offspring, the serum levels of PEDV-specific IgG and IgA antibodies were measured in both newborn (at birth) and weaned piglets (4 weeks of age). The evaluation encompassed both the efficacy of maternally derived antibody transfer and the subsequent decay kinetics characteristic of passive immunity. The analysis utilized a repeated measures ANCOVA, adjusting the maternal antibody level of the corresponding sows to provide a more precise assessment of passively acquired immunity. The results are detailed in Fig. 6. Anti-PEDV IgG and IgA responses in piglet sera were examined via commercial and in-house COE- and PEDVSME-based ELISA kits. Notably, the IgG and IgA levels were robust in newborn piglets but markedly decreased close to baseline levels (mean OD450 < 0.2) in weaned piglets in all groups (Fig. 6A,

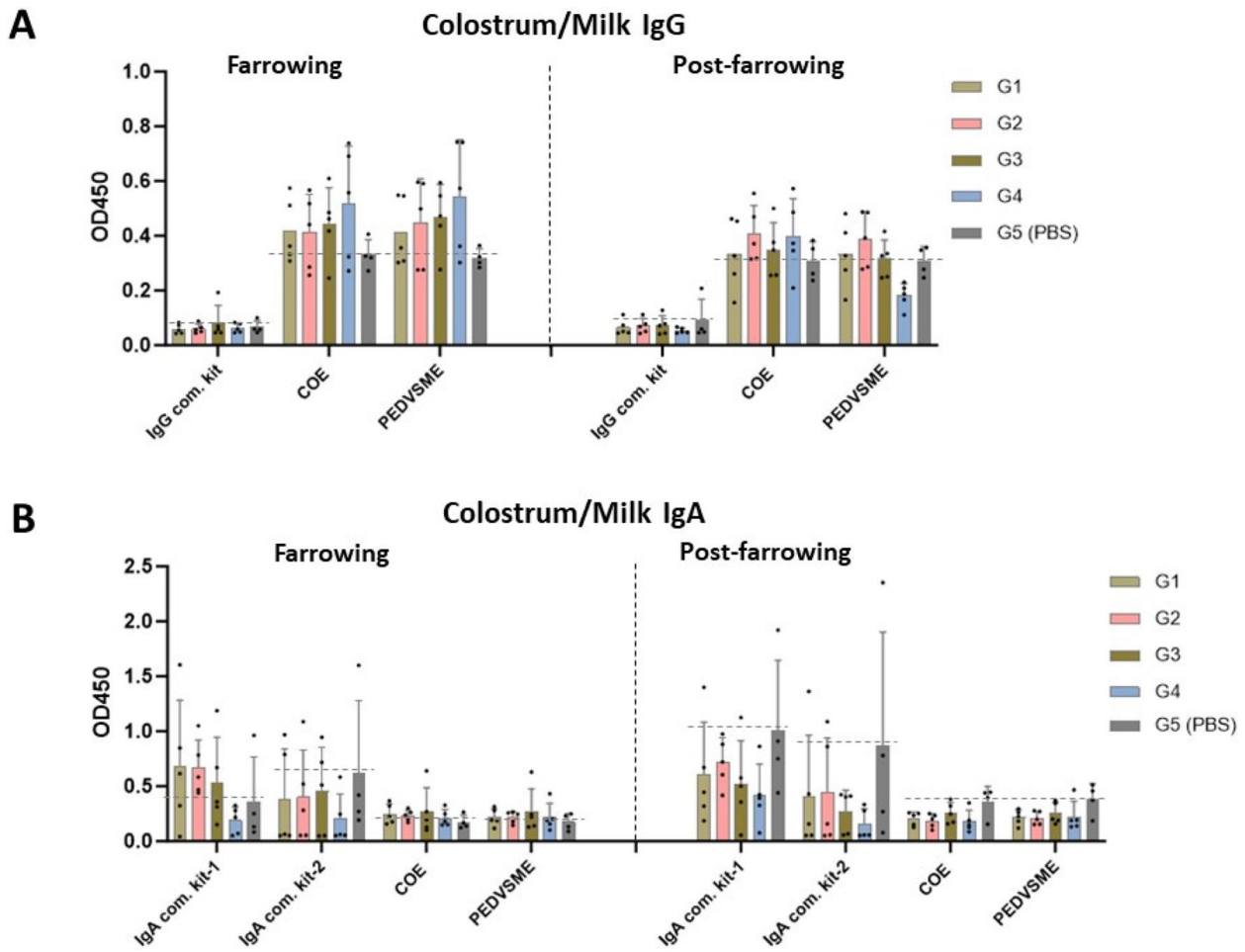

Fig. 5 IgG and IgA responses in sow mammary secretions following immunization. **A** IgG and **B** IgA antibody levels were measured at two time points: farrowing (colostrum) and 5–9 days postfarrowing (milk). G1-G4 represent the immunization groups as follows: rAd. PEDVSME (G1), rAd. PEDVSME-OmpH (G2), rAd. PEDVSME-CTB (G3), rAd. PEDVSME-GI (G4), and the control group (G5)

C), indicating the expected decay of maternally derived antibodies. This finding was also consistent with the decrease in antibody levels in milk after farrowing.

Furthermore, the sow's antibody level, a significant predictor of the piglet's antibody level, was included as a covariate in the model. This finding indicates a direct positive correlation where sows with higher immunity transfer greater levels of antibodies to their offspring. The analysis of adjusted mean IgG levels revealed no significant main effect for the vaccine group, nor was there a significant interaction effect between the vaccine group and time (Fig. 6B). This overall result suggests that,

(See figure on next page.)

Fig. 6 Serum IgG and IgA responses in piglet sera. Piglets were born to sows immunized with the following rAds: rAd. PEDVSME (G1), rAd. PEDVSME-OmpH (G2), rAd. PEDVSME-CTB (G3), rAd. PEDVSME-GI (G4) and the control group (G5). Panels **A** and **C** display the distributions of raw OD450 values for IgG and IgA, respectively, as scatter dot plots, with bars representing the means ± SDs. Panels **B** and **D** illustrate the adjusted marginal means with standard errors (bars) of the IgG and IgA levels, respectively, derived from statistical modeling *via* repeated measures analysis of covariance (ANCOVA), with the corresponding sow's antibody level as a covariate. Statistical significance is denoted as follows: between-group comparisons: lowercase letters (a, b) indicate significant differences between study groups within a single time point. Groups not sharing a common letter are significantly different ($P < 0.05$, Tukey's HSD). Within-group comparisons: asterisks indicate a significant difference between the newborn and weaned periods within the same study group (*$P < 0.05$; ** $P < 0.01$; *** $P < 0.001$)

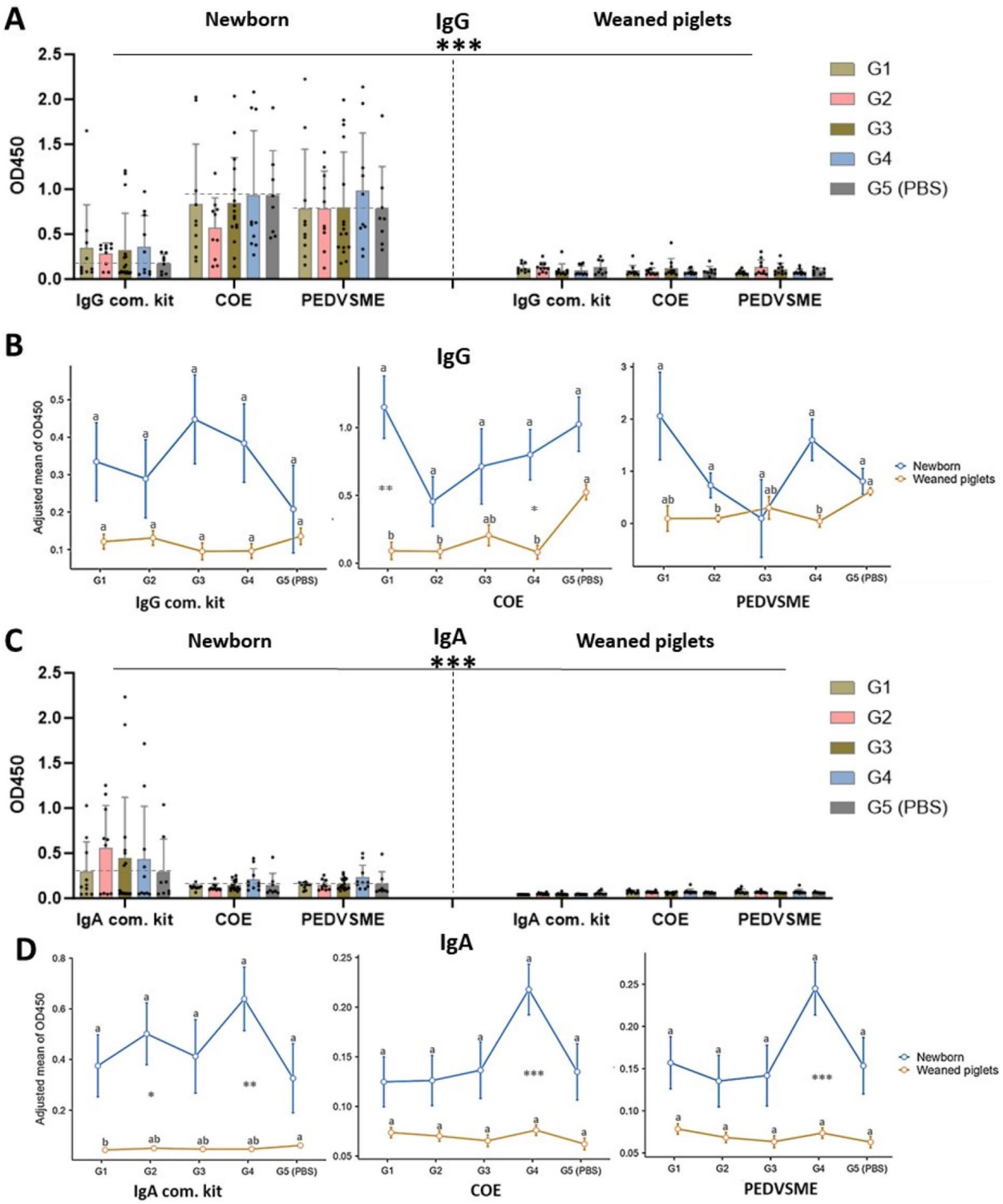

Fig. 6 (See legend on previous page.)

while passive antibody transfer occurred, no single vaccine construct was statistically superior to the others, and none of the vaccines altered the natural decay rate of these maternally derived antibodies. At the weaning time point, no significant differences in IgG levels were detected among the study groups.

The levels of passively acquired serum IgA were also assessed at birth and weaning (Fig. 6C and 6D). At birth, the adjusted mean IgA level had no significant effect on the vaccine group (Fig. 6D). A significant reduction in serum IgA levels was observed from birth to weaning, particularly in the groups with the highest initial titers, rAd. PEDVSME-OmpH (G2) and rAd. PEDVSME-GI (G4) ($P < 0.01$ or $P < 0.001$). At the weaning time point, IgA levels decreased to a minimal baseline in all piglets, and no significant differences were detected among any of the study groups (OD450 < 0.1).

Taken together, these findings demonstrate effective maternal transfer of vaccine-induced antibodies, both IgG and IgA, to newborn piglets. However, both isotypes dramatically declined after four weeks (weaning period), suggesting the potential need for active immunization for continued protection against PEDV infection.

Characterization of the antibody response to PEDVSME B-cell epitopes

As our vaccine was designed on the basis of the epitope-based vaccine concept, here, we studied the antibody response against a panel of B-cell epitopes, which are included in the PEDVSME immunogen. (Fig. 7). These include epitopes in the COE region and reported neutralizing epitopes: S1B-8 (peptide M), S1B-19 (S1D5 and S1D6), and S2B-33 (2C10). In sow sera, the differential IgG response against all B-cell epitopes moderately increased after the second dose in all groups, and there were no significant differences between the immunized groups (G1–G4) and the control group (G5) (Fig. 7A). Interestingly, analysis of colostrum samples revealed higher IgG than IgA responses (Fig. 7B, 7C). The IgG levels in colostrum ranged from 0.5–1.0 across most epitopes. Colostrum IgG responses against all epitopes in the immunized groups, particularly the rAd. PEDVSME (G1) and rAd. PEDVSME-CTB (G3), which was consistently greater than that in the control group (G5) (Fig. 7B). Somewhat unexpectedly, the PBS control group presented higher IgA levels against several epitopes than the immunized groups did (Fig. 7C). However, several epitopes in the COE region, including COE-2, COE-3, and S1B-17, displayed greater IgA responses in the immunized groups (G1–G4) than in the control group (G5). This study suggested that all B-cell epitopes composing the PEDVSME immunogen vectored by adenovirus were capable of inducing antibody responses in gestating sows. However, most epitopes preferentially induced and secreted PEDV-specific IgG rather than IgA in colostrum/milk.

Neutralizing antibodies in sow milk and piglets

To determine the protective capacity of vaccine-induced humoral immunity against PEDV, both active and passive, we examined neutralizing antibody titers via a neutralization assay in the sow sera, colostrum/milk, and sera of offspring (Fig. 8). In immunized sows, neutralizing antibody titers were monitored at three time points: preimmunization, post first dose, and post second dose. All groups receiving vaccines (G1–G4) developed neutralizing antibodies following immunization, with titers ranging from 4–16 after the second dose, whereas the control group (G5) maintained titers below 1. Compared with sera from the preimmunization groups, sera from all four immunized groups presented increased neutralizing titers after the second dose (Fig. 8A**,** left panel). ANCOVA revealed that a sow's preexisting neutralizing titer was a highly significant predictor of its postvaccination response ($P < 0.001$), indicating that baseline immunity strongly influenced the magnitude of the antibody response. On the basis of the adjusted mean titers (Fig. 8A, right panel), the rAd. PEDVSME-CTB (G3) exhibited the most robust systemic neutralizing antibody response, followed by rAd. PEDVSME-OmpH (G2), after the first and second immunizations. However, the analysis revealed no statistically significant differences in neutralizing titers among the four immunized groups (G1-G4) and the control group (G5). Furthermore, the raw data suggested an increase in titer from the first to second doses; however, this booster effect (rate of increase) was not statistically significant among the different groups.

Analysis of neutralizing antibodies in milk samples collected at the farrowing (colostrum) and postfarrowing (milk) stages revealed successful transfer of PEDV-specific neutralizing antibodies to colostrum/milk. All immunized groups (G1–G4), including the control group (G5), presented mean neutralizing titers in colostrum and milk ranging from 8–16 (Fig. 8B, left panel). This suggests a natural infection in the control group. However, the neutralizing titers in the colostrum of all four immunized groups (G1–G4) were greater than those in the control group (G5). Importantly, these neutralizing antibodies persisted in milk throughout the postfarrowing period, although with some decline in titers that were comparable to those of the control group (G5). For further analysis, the neutralizing titer from sow sera was used as a covariate (Fig. 8B, right panel). The statistical analysis indicated that the sow's serum antibody level was a significant factor influencing the neutralizing titer found in milk. However, the model did not result in any statistically significant differences in milk neutralizing titers, either among the different experimental groups or between the farrowing and postfarrowing periods. These

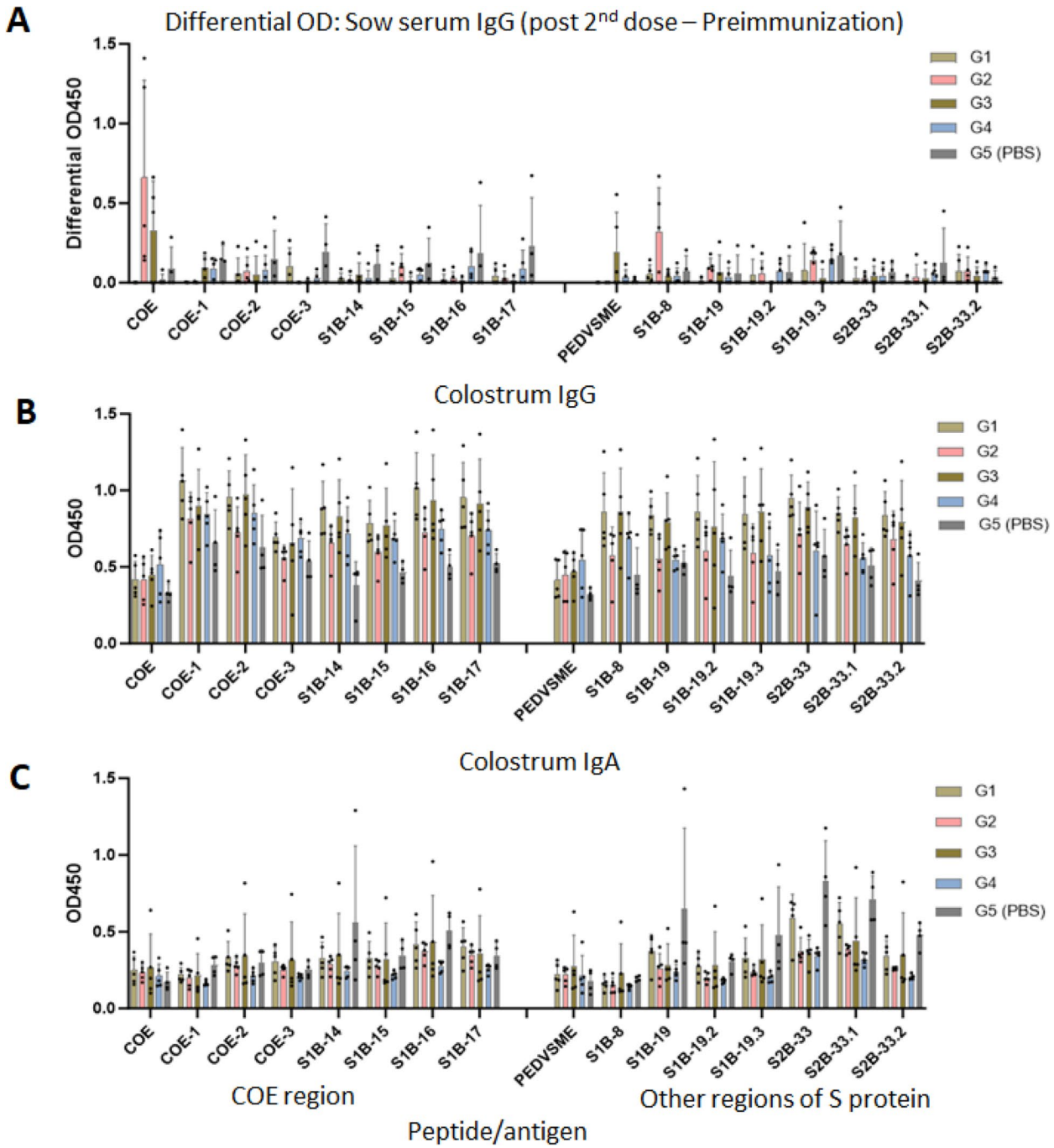

Fig. 7 IgG and IgA responses to linear B-cell epitopes in the PEDVSME immunogen. **A** Differential IgG responses in sow sera against linear B-cell epitopes. **B** IgG responses and **C** IgA responses against linear B-cell epitopes in colostrum samples collected from sows at farrowing. The data are presented as the means ± SDs. The B-cell epitopes are grouped as epitopes located within and outside the COE region. G1-G4 represent the immunization groups as follows: rAd. PEDVSME (G1), rAd. PEDVSME-OmpH (G2), rAd. PEDVSME-CTB (G3), rAd. PEDVSME-GI (G4) and the control group (G5)

findings suggest that none of the vaccine constructs were superior to the others in promoting this transfer and that the levels remained stable during the measured lactation periods.

Examination of piglet sera confirmed successful transfer of maternal antibodies to offspring. Compared with control piglets, newborn piglets from vaccinated sows (G1–G4) presented higher neutralizing titers ranging

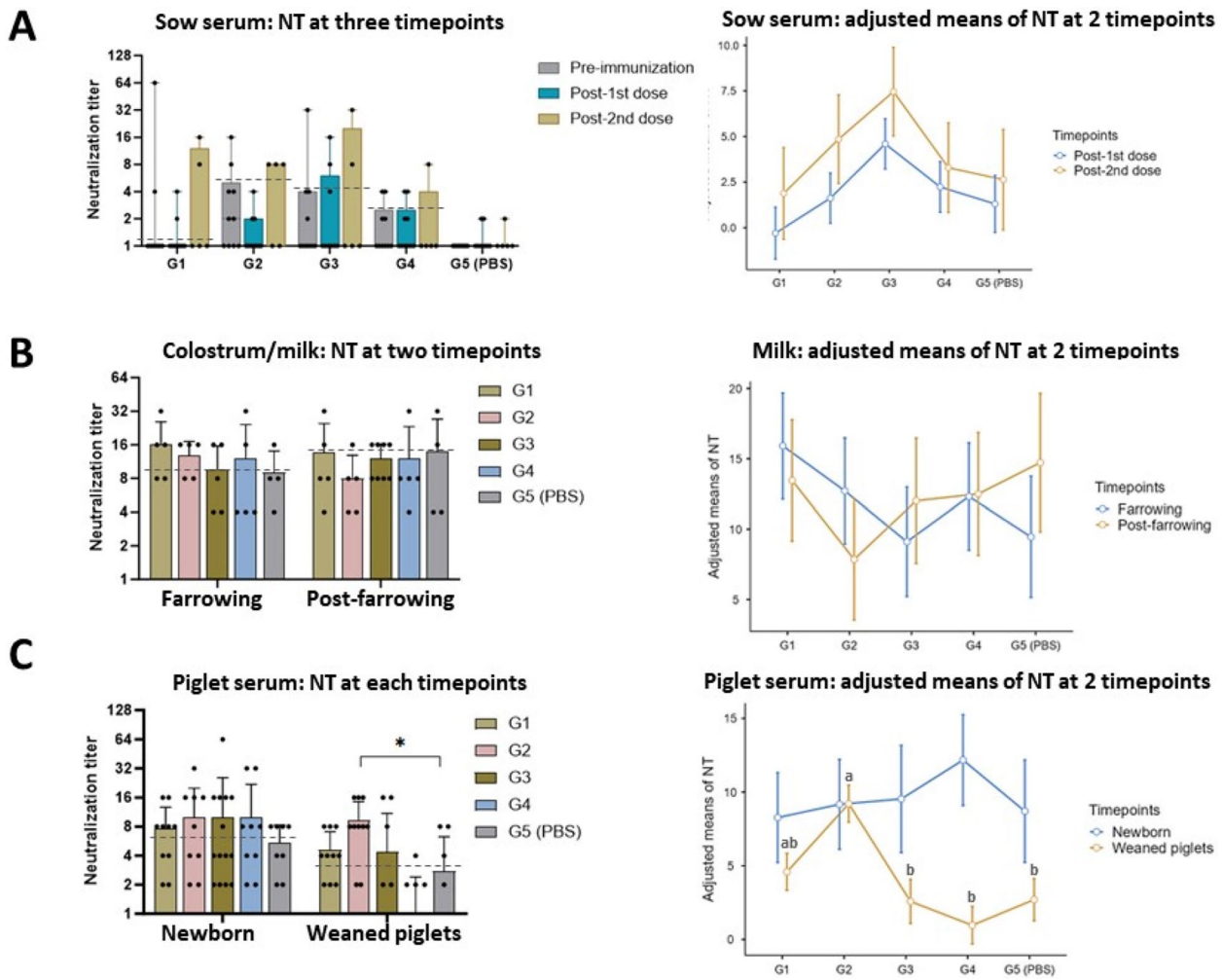

Fig. 8 PEDV-specific neutralizing antibody responses in sow sera, colostrum/milk, and piglet sera following immunization. Sows were immunized with different recombinant adenovirus-based vaccines: rAd. PEDVSME (G1), rAd. PEDVSME-OmpH (G2), rAd. PEDVSME-CTB (G3), rAd. PEDVSME-GI (G4) and the control group (G5). For each panel, the left graph displays individual animal neutralizing titers (dots) with the mean ± SD, whereas the graph on the right displays the estimated marginal mean ± standard error of the mean (SEM) derived from a repeated measures ANCOVA model. **A** Sow serum neutralizing titers. The left panel shows the titers before immunization, after the 1 st dose, and after the 2nd dose. The right panel shows adjusted means after the 1 st and 2nd doses, with the preimmunization titer used as a covariate. **B** Colostrum/milk NT titers. The left panel shows titers at farrowing and postfarrowing. The right panel shows adjusted means for the same time points, with the post-2nd-dose sow serum titer used as a covariate. **C** Piglet serum neutralizing titers. The left panel shows the titers in newborn and weaned piglets. The right panel shows adjusted means for these time points, with the post-2nd-dose sow serum titer used as a covariate. In this panel, groups at the weaned stage not sharing a common letter (a, b) are significantly different according to Tukey's post hoc test ($P < 0.05$)

from 16–64 (Fig. 8C, left panel). While titers generally decreased with weaning age (4 weeks of age). Covariant analysis revealed that the sow's systemic neutralizing titer was a highly significant covariate, confirming that the level of maternal immunity is a crucial determinant of the neutralizing titer in offspring (Fig. 8C, right panel). The overall ANCOVA model revealed no significant main effect for the vaccine group and no significant interaction between group and time. The titers in the other groups were not significantly different from those in the control group (G5). This suggests that the vaccine G2 conferred a more durable passive neutralizing antibody response that persisted at a higher level until weaning.

Correlations between antibody responses and neutralizing titers

Furthermore, we studied the relationship between the antibody response (OD450 values of both IgG and IgA) and neutralizing titer in the serum and colostrum/milk samples. The analysis was performed by pairing

neutralizing titers with the IgG and IgA levels obtained from different ELISA tests: commercial kits and in-house COE- and PEDVSME-based ELISA. Interestingly, we observed the strongest correlation between the IgG levels and neutralizing titers in colostrum and milk samples. The colostrum IgGs tested by the commercial IgG ELISA kit demonstrated the highest correlation coefficient ($R^2 = 0.232$), whereas the IgGs tested by our in-house ELISAs based on COE and PEDVSME antigens also showed meaningful correlations ($R^2 = 0.105$ and 0.086, respectively) in the colostrum samples (Fig. 9A). Surprisingly, colostrum IgA levels obtained from all four ELISA tests were less correlated with neutralizing titers than were colostrum IgG levels. The IgA concentrations obtained from commercial kits demonstrated correlation coefficients (R^2) of 0.095 and 0.141 for the Innoceleris and IDEXX kits, respectively. Compared with those obtained from the commercial kit, the IgA responses obtained from the COE- and PEDVSME-based ELISAs were less correlated with neutralizing activity, with $R^2 < 0.03$ in both colostrum and milk samples (Fig. 9B).

To investigate the relationship between maternal immunity and passive immunity in piglets, we analyzed maternal serum IgG levels and their relationship with neutralizing titers in newborn piglets. The commercial ELISA kit showed the most promising correlation ($R^2 = 0.187$), while the correlations with in-house ELISAs were lower ($R^2 < 0.04$). Taken together, these findings highlight the potential utility of these commercial and in-house IgG ELISA tests for evaluating and predicting protective immunity during the critical period of antibody transfer to newborn piglets. By considering both antibody levels and correlations with neutralizing capacity, in-house COE- and PEDVSME-based IgG ELISAs offer greater potential for indicating protective immunity in colostrum and milk than commercial IgG and IgA ELISAs do. However, commercial IgG ELISA for detecting IgG in sow serum may also be beneficial in predicting passive immunity from sows to piglets.

Cell-mediated immune response in vaccinated sows

In addition to investigating humoral immunity, we investigated the cell-mediated immune responses induced by our PEDV vaccines. An IFN-γ ELISPOT assay was performed with PBMCs isolated from sows at two time points: preimmunization and 3 weeks after the first dose. PBMCs were cultured under two conditions: an unstimulated control (no peptide) and a mixture of PEDV peptide (composed of S1B-8, S1B-15, S1B-16, S1B-17, S1B-19, and S2B-33). Analysis of preimmunized samples revealed substantial baseline IFN-γ responses ranging from 10–10,000 spots per million cells under both unstimulated and peptide-stimulated conditions (Fig. 10A). The presence of spots in unstimulated wells at levels comparable to those in peptide-stimulated wells suggests preexisting T-cell immunity in these animals prior to vaccination.

To assess vaccine-induced T-cell responses, we calculated the differential spot counts (subtraction of spot number at postimmunization from that at preimmunization) for each animal (Fig. 10B). Enhanced PEDV-specific IFN-γ responses were observed in at least one animal from groups G1–G4, indicating an expansion of the PEDV-specific T-cell populations following immunization. Notably, the groups receiving rAd. PEDVSME-CTB (G3) and rAd. PEDVSME-GI (G4) demonstrated the highest frequency of responders, suggesting the strong potency of these vaccines at inducing T-cell-mediated immunity. These results demonstrate the potential of the mucosal adjuvants CTB and GI in enhancing the cellular immune response when they are incorporated into an adenoviral vector.

Observation of vaccine safety issues

To address vaccine safety concerns, the immunized sows (n = 50) and their piglets were kept under close observation for possible adverse reactions from the vaccine. Within 24 h of vaccination, there were no reports of either local reactions (e.g., pain, swelling, redness, abscess formation) or systemic reactions (e.g., hypersensitivity, anaphylaxis, pyrexia, anorexia) in any of the studied animals. Similarly, no abnormalities were reported at 48 h, 72 h, or 7 days postvaccination. None of the immunized sows or piglets presented clinical signs of PED infection (e.g., watery diarrhea, vomiting, dehydration, and anorexia). RT–qPCR confirmed that there was no PEDV particle shedding in the feces of any of the immunized

(See figure on next page.)

Fig. 9 Correlation between neutralizing antibody titers and antibody responses. The data represent an individual sample. **A** Correlations between neutralizing titers and IgG levels (measured as the OD450) in colostrum (top row) and postfarrowing milk (bottom row) determined via three different commercial ELISA kits. **B** Correlations between neutralizing titers and IgA levels (OD450) in colostrum (top row) and postfarrowing milk (bottom row) determined via four different commercial ELISA kits**.** **C** Correlation between sow serum IgG levels (OD450) at farrowing and the resulting serum neutralizing titers in newborn piglets. The coefficient of determination ($R.^2$) is shown for each correlation analysis, indicating the strength of the relationship between the neutralizing antibody titer (x-axis) and the antibody level measured by ELISA (y-axis)

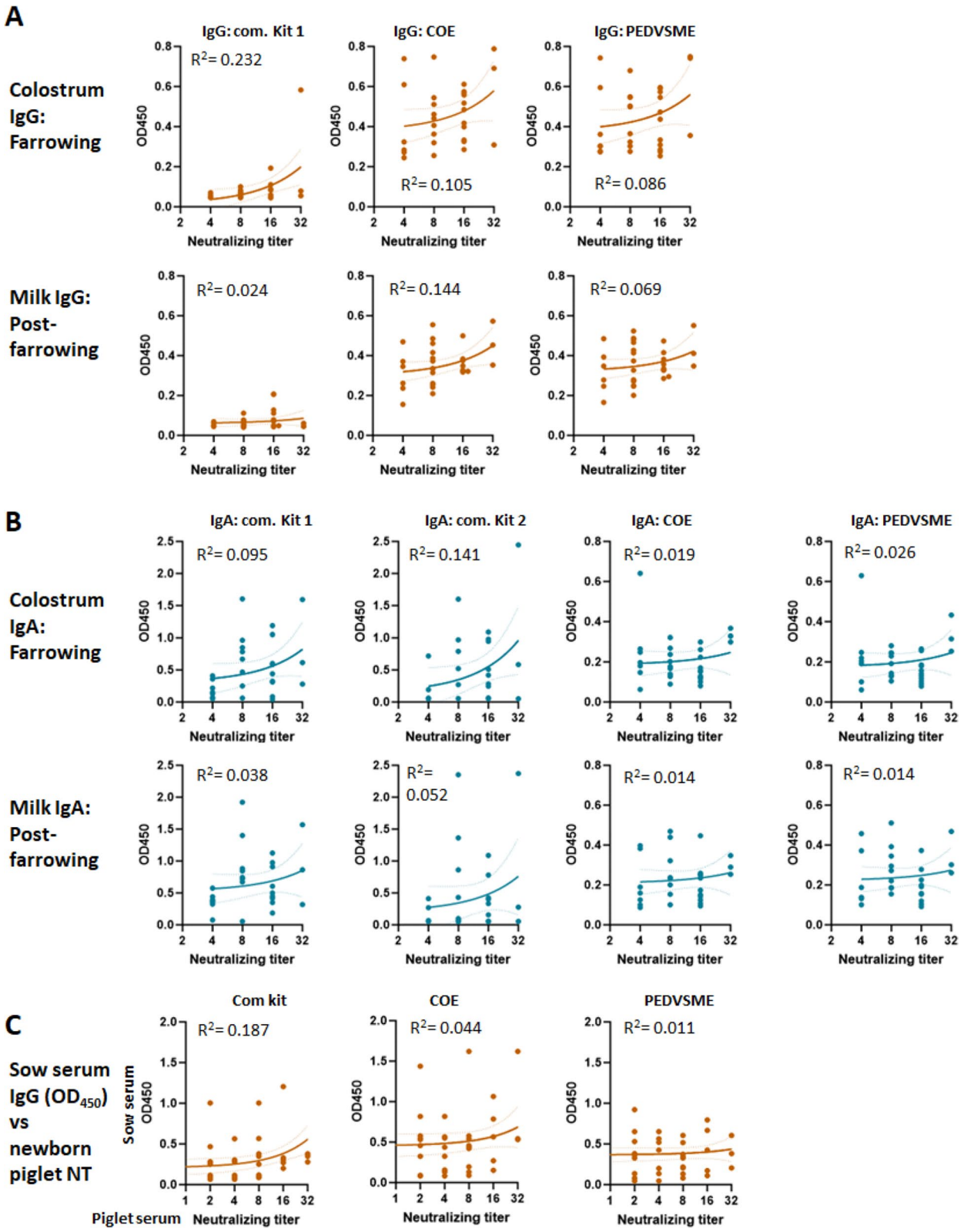

Fig. 9 (See legend on previous page.)

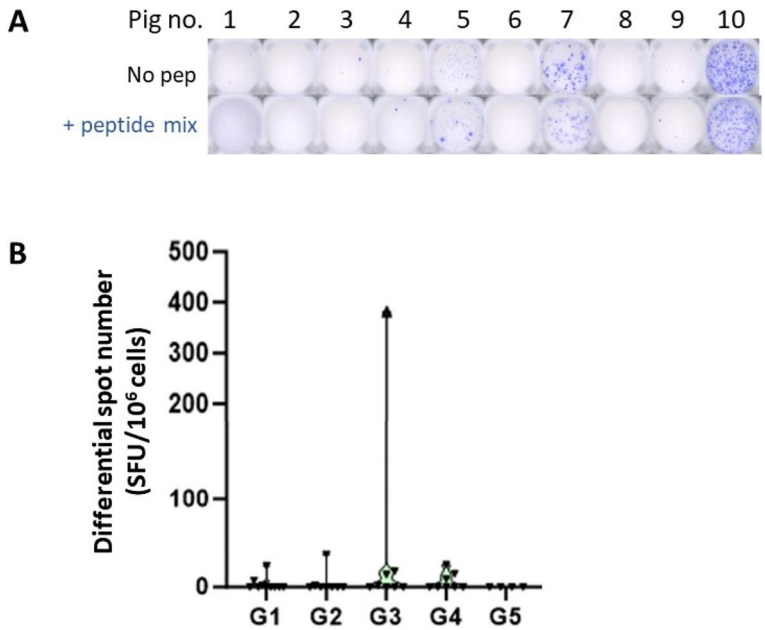

Fig. 10 Cellular immune response in immunized sows. The PEDV-specific T-cell response was examined via the IFN-γ ELISPOT assay. **A** Representative ELISPOT wells showing IFN-γ spots from without peptide stimulation (No pep) or with PEDV peptide mixture stimulation (+ peptide mixture). **B** Differential IFN-γ spot numbers between post- and prevaccination samples from individual sows across different immunization groups. The y-axis shows the number of spot-forming units (SFUs) per million cells. Experimental groups: rAd. PEDVSME (G1), rAd. PEDVSME-OmpH (G2), rAd. PEDVSME-CTB (G3), rAd. PEDVSME-GI (G4), and the control group (G5)

sows. Hence, these findings implied that our vaccine prototypes, rAd. PEDVSMEs linked with mucosal adjuvants did not induce adverse reactions in the studied animals.

Discussion

As PEDV primarily infects and replicates in intestinal epithelial cells, the induction of robust mucosal immunity is critical for effective protection. While recombinant adenovirus (rAd) vectors are recognized platforms for developing PEDV vaccines (Liu et al. 2019; Do et al. 2020), strategies to optimize their immunogenicity, particularly for mucosal responses, continue to evolve. Previous studies have demonstrated that rAd vectors expressing the full-length PEDV spike (S) protein can elicit protective immunity (Liu et al. 2019). Furthermore, Liu et al. reported that rAd expressing the PEDV S protein successfully elicited a strong PEDV-specific immune response in 4-week-old pigs and conferred protection against virulent PEDV strains (Liu et al. 2019). Similarly, Do et al. demonstrated that Ad-LTB-COE, which incorporates adjuvants such as heat-labile enterotoxin B (LTB) in the COE region, increases immune responses to rAd-delivered PEDV antigens (Do et al. 2020).

However, there remains an opportunity to further enhance the mucosal immunogenicity of these vaccines through advanced antigen design and the exploration of alternative adjuvants. To address this, we developed a novel multiepitope immunogen, PEDVSME, and engineered its delivery via rAd vectors in tandem with powerful mucosal adjuvants: OmpH, CTB, or a fusion cytokine (GI). Our work distinguishes itself from previous efforts by creating a chimeric immunogen and directly comparing of these specific adjuvant fusions within a rAd backbone. The successful expression of these complex immunogens in both the HEK293 and PK-15 cell lines, as demonstrated here, serves as the foundation for this advanced vaccine design, paving the way for evaluating its superior potential in vivo. Our results are consistent with those of previous studies demonstrating efficient transgene delivery via human adenovirus type 5 vectors in swine (Torres et al. 1996; Song et al. 2024).

The primary objective of this study was to evaluate the immunogenicity of four different recombinant adenovirus (rAd)-based PEDV vaccine candidates in gestating sows, with particular emphasis on systemic immunity, lactogenic antibody transfer, and the critical endpoint of passive immunity in piglets. Upon immunization, all vaccine candidates induced immune responses superior to those in the control group (G5). Crucially, the choice of adjuvant was found to critically influence the resulting immune profile. The rAd. The PEDVSME-CTB vaccine (G3) was superior for inducing systemic IgG responses in sows. In contrast, the vaccine G2 was the most effective eliciting systemic IgA and, most importantly, maintaining

significantly higher neutralizing antibody titers in piglets throughout the critical weaning period. While the study confirmed the successful maternal transfer of antibodies, it also highlighted their predictable decay, underscoring the necessity for vaccine strategies that confer potent and durable passive protection.

Our findings demonstrated that fusing different adjuvants to an identical immunogen can polarize the systemic immune response in sows. A key finding was the superior ability of the vaccine G3 expressing the highest systemic IgG levels and robust neutralizing antibody (NT) titers in immunized sows. This finding aligns with the established role of CTB as a potent mucosal adjuvant that also strongly promotes systemic Th2-type responses, which are essential for high-titer IgG production (Pacheco et al. 2000; Bailey et al. 2017; Symowski and Voehringer 2019; Bernstein et al. 2023). CTB functions as a potent mucosal adjuvant that also strongly promotes systemic Th2-type responses through its ability to bind GM1 ganglioside receptors on antigen-presenting cells, leading to increased antigen presentation and subsequent B-cell activation (Pacheco et al. 2000; Bailey et al. 2017; Symowski and Voehringer 2019). This mechanism explains the pronounced increase in IgG production observed in the vaccine G3. Th2 responses are characterized by the secretion of IL-4 and IL-13, which drive B-cell differentiation toward IgG-producing plasma cells (Bernstein et al. 2023).

OmpH, an outer membrane protein from *Yersinia enterocolitica*, specifically targets M cells within gut-associated lymphoid tissues (GALTs), facilitating efficient antigen sampling and presentation (Grassl et al. 2003; Kim et al. 2011; Rios et al. 2016; Sato and Pascual 2020). The resulting immune response appears to favor the generation of IgA-committed B cells, which subsequently home to mucosal sites and contribute to both systemic and secretory IgA production (Rios et al. 2016). This characteristic is particularly desirable for vaccines targeting mucosal pathogens such as PEDV. This observation is consistent with findings from Li et al., that the incorporation of OmpH into a PED subunit vaccine significantly improved specific humoral and mucosal immunity in mice (Li et al. 2018).

The successful secretion of antibodies into mammary secretions represents a crucial component of the maternal vaccination strategy. All vaccinated groups demonstrated the ability to transfer antibodies, particularly IgG, into colostrum at levels significantly greater than those in the control group. This finding validates the fundamental premise that intramuscular vaccination of gestating sows can effectively stimulate antibody production and subsequent transfer to mammary secretions. With respect to lactogenic immunity, our results revealed efficient transfer of vaccine-induced antibodies to mammary secretions, with the strongest IgG responses in the rAd. PEDVSME-GI group (G4). This observation is crucial since maternal antibody transfer is essential for protecting newborn piglets against PEDV infection. The enhanced lactogenic immunity observed in the vaccine G4 can be attributed to the immunomodulatory properties of GM-CSF and IL-4. GM-CSF plays a crucial role in the maturation and differentiation of dendritic cells, which are essential for initiating robust immune responses (Conti and Gessani 2008; Van De Laar et al. 2012). IL-4 has been demonstrated to play a critical role in the progression of humoral immune responses and the differentiation and maturation processes of B lymphocytes (Nelms et al. 1999; Vazquez et al. 2015). Furthermore, IL-4 has been shown to upregulate the expression of polymeric immunoglobulin receptor (pIgR) in mammary epithelial cells, facilitating the transport of immunoglobulins into milk (Salmon et al. 2009; Wei and Wang 2021). The combination of GM-CSF and IL-4 in our vaccine construct appears to create an optimal immunological environment for increased antibody production and transfer.

Interestingly, while IgA is generally recognized as a main isotype in lactogenic immunity, we found that the IgA levels in colostrum/milk were lower than the IgG levels measured by our in-house COE- and PEVSME-based ELISAs. However, it is also important to consider that this finding could be influenced by differences in the analytical sensitivity or dynamic range of the in-house ELISAs used for detecting porcine IgG versus IgA. While both ELISAs were optimized and validated for their respective targets, a direct comparative analysis of their sensitivity limits was not formally conducted as part of this study. Therefore, this potential methodological aspect should be considered when interpreting the relative magnitudes of IgG and IgA responses. The results of colostrum IgG and IgA responses against individual B-cell epitopes also confirmed this pattern of response. A study by Song et al. reported high IgG levels in colostrum, but these levels were still slightly lower than the IgA levels (Song et al. 2023). This may suggest that systemic immunity contributes to lactogenic immunity. Moreover, incorporating mucosal adjuvants (OmpH) enhanced IgA secretion in milk but did not alter the pattern of response (IgG > IgA). In addition to the properties of the immunogen, the vaccine platform may also contribute to the increased level of IgG in milk. As our vaccines are vectored by adenovirus and given to sows *via* intramuscular injection, this regimen may be more effective at inducing systemic than mucosal immune responses.

Although it has been shown that PEDV-specific IgA in colostrum was more strongly correlated with neutralizing

antibody activity than was IgG (Song et al. 2023), our correlation analysis indicated that the IgG response in colostrum/milk was directly correlated with the neutralizing titer, with a correlation coefficient greater than that of the IgG level measured our in-house ELISA. Our findings suggest that colostrum/milk IgG contributes to the neutralization of PEDV, thus providing protective immunity to piglets via passive transfer. Moreover, the results also highlight the possibility of using our in-house ELISA as an alternative kit to assess the antibody response in colostrum/milk, which can guide neutralizing capacity on the basis of the IgG level.

Moreover, neutralizing antibody analysis revealed that all four vaccines effectively induced neutralizing antibodies in sow sera, with titers ranging from 4–16 after the second dose. Although these neutralizing titers are modest, the study by Chang et al. indicated that these titers were sufficient to confer passive protection to piglets (Chang et al. 2019). Importantly, correlation analyses demonstrated a direct association between sows' IgG and neutralizing titers in piglets. These findings suggest that the level of the antibody response in sows could indicate protective immunity passively transferred to piglets. Furthermore, Song et al. reported that maternal immunization with an adenovirus expressing the PEDV S protein (Ad-PEDV-S) increased piglet survival and reduced clinical symptoms and viral loads in piglets after challenge (Song et al. 2024). While our study confirmed successful transfer of antibodies from vaccinated sows to their offspring, we also observed the natural and expected decay of these maternal antibodies over time, emphasizing the critical importance of both antibody quantity and durability. The most significant finding of our study was that rAd. PEDVSME-OmpH (G2) was the only vaccine construct that resulted in significantly higher neutralizing antibody titers in piglets at the critical weaning period the control group (G5). This observation is particularly remarkable because, while rAd. PEDVSME-CTB (G3) produced higher systemic IgG and neutralizing titers in the sows themselves; the antibodies conferred by rAd. PEDVSME-OmpH (G2) demonstrated superior durability in piglets. We hypothesize that the OmpH adjuvant may lead to the production of higher-affinity antibodies or antibodies with an extended half-life. Alternatively, the strong systemic IgA response driven by the OmpH may translate to a more sustained or effective transfer of neutralizing capacity via milk. This result firmly establishes that the quality and durability of passive immunity, rather than the peak maternal titer alone, are the most important metrics for PEDV vaccine efficacy.

The assessment of cell-mediated immunity in our study presents both promising results and significant interpretive challenges. The detection of substantial preexisting T-cell responses in sows represents a major limitation that complicates the interpretation of vaccine-induced cellular immunity. This phenomenon is common in outbred animal populations with previous pathogen exposure and highlights the complexity of evaluating vaccine efficacy under field conditions. Despite this limitation, our results demonstrate that adjuvanted vaccines, particularly rAd. PEDVSME-CTB (G3) and rAd. PEDVSME-GI (G4) appeared to increase cell-mediated immunity in several animals, as evidenced by increased IFN-γ production following stimulation with PEDV antigens. This finding demonstrates the potential of the adenoviral vector platform, when enhanced with appropriate adjuvants, to stimulate the cellular arm of adaptive immunity. The ability to induce both humoral and cellular responses is particularly valuable for antiviral vaccines, as cellular immunity plays a crucial role in viral clearance and long-term immunological memory. The observed effects are consistent with the known functions of CTB and GI fusion cytokines in promoting dendritic cells (DCs) maturation and T-cell activation (Van De Laar et al. 2012; He et al. 2022; Meas et al. 2024). In addition, He et al. demonstrated that CTB promotes dendritic cell maturation and enhances antigen presentation, leading to improved T-cell responses (He et al. 2022). Additionally, Wang et al. reported that CTB can activate both CD4+and CD8+T cells, contributing to long-lasting immune memory (Wang et al. 2003). The GI adjuvant, comprising GM-CSF and IL-4, likely enhances cellular immunity through complementary mechanisms: GM-CSF promotes dendritic cell recruitment and maturation (Conti and Gessani 2008; Van De Laar et al. 2012), whereas IL-4 supports T helper cell differentiation and memory T-cell formation (Fairfax et al. 2015). A recent study by Meas et al. further confirmed that coexpressing GM-CSF and IL-4 with a vaccine antigen significantly enhanced humoral and cellular immune responses in a PCV vaccine model (Meas et al. 2024).

To ensure a balanced interpretation of our findings, several limitations must be acknowledged. First and most critically, this study did not include a viral challenge phase; therefore, while we measured strong immune correlates of protection, particularly NT antibodies, we did not definitively confirm protection against live PEDV infection in piglets. While neutralizing antibodies serve as well-established correlates of protection for many viral diseases, definitive confirmation of their protective efficacy requires direct challenge studies. This represents the most important next step in the development of our lead vaccine candidate. Second, the high baseline immunity observed in our study population, while reflecting realistic field conditions, made it more difficult to discern the true magnitude of vaccine responses. This was

particularly evident in the assessment of cell-mediated immunity and may have contributed to the detection of neutralizing antibodies in some control animals. The presence of preexisting immunity likely reduced the statistical power to detect differences between groups and may have masked more substantial vaccine effects. Third, while several key differences were statistically significant, the relatively small sample size (n = 10/group) and high individual variation may have limited the statistical power to detect more subtle differences between vaccine formulations. Finally, the GI fusion protein was not detectable by Western blot, although its expression was confirmed via immunofluorescence.

Conclusion

Our study demonstrated the ability of adenovirus-based PEDV vaccines to elicit immune responses in sows and provide lactogenic immunity to piglets. The novel immunogen PEDVSME and its derivatives modified with the mucosal adjuvants OmpH, CTB, and GI in an adenoviral vector induced PEDV-specific antibodies in gestating sows, which could be transferred to newborn piglets and sustained to the weaning period. Furthermore, no adverse vaccine reactions were reported during the monitoring period. Taken together, these findings indicate that the rAd. PEDVSME-OmpH (G2) is considered the lead candidate for further study. The superiority of this vaccine lies not only in its ability to induce a strong maternal immune response but also in its capacity to provide the most durable passive neutralizing immunity in piglets at weaning.

The successful induction of both humoral and cellular immunity, coupled with efficient maternal antibody transfer, suggests that these vaccine candidates merit further development as alternative vaccines against PEDV. Nonetheless, further optimization is needed to maximize the efficacy of these Ad-based vaccines, such as the use of mucosal delivery methods and other mucosal adjuvants to increase IgA production while sustaining robust IgG responses. Furthermore, long-term studies assessing their protective efficacy against live PEDV challenge in piglets will be crucial to validate their practical applicability in the field.

Methods

Generation of recombinant adenovirus

Recombinant adenovirus vectors were constructed via the Gateway® cloning system (Walhout et al. 2000). First, the immunogen genes were cloned and inserted into the pENTR11 dual selection vector (Gateway™ pENTR™ 11; Invitrogen, Thermo Fisher Scientific, Carlsbad, CA, USA) to generate entry clones. The immunogen expression cassettes were subsequently transferred into the adenoviral destination vector pAd/CMV/V5-DEST (Invitrogen) via site-specific LR recombination via Clonase® II enzyme mix (Invitrogen). The recombinant plasmids were propagated in *Escherichia coli* DH5α. The desired recombinant clones were further screened by rapid size screening. The recombinant clones were subsequently confirmed for the existence of the target gene via restriction enzyme digestion, and the nucleotide sequence was subsequently confirmed via automated DNA sequencing. For virus production, the vector pAd/CMV/V5-DEST harboring immunogen genes was linearized with *Pac*I (New England Biolabs, Ipswich, MA, USA) and transfected into human embryonic kidney cells (HEK293A). The cells were maintained in Dulbecco's modified Eagle's medium (DMEM; HiMedia Laboratories, Mumbai, Maharashtra, India) supplemented with 10% fetal bovine serum (FBS, HiMedia Laboratories Private Limited) and 1% penicillin–streptomycin (Gibco, Thermo Fisher Scientific, Grand Island, NE, USA) in a 5% CO_2 incubator at 37°C. Transfections were performed at 80–90% cell confluency in 6-well plates, followed by incubation for 7–10 d to allow for viral particle production. The recombinant adenovirus was harvested and amplified through successive passages in HEK293A cells. The final viral preparations were purified via cesium chloride (CsCl) density gradient ultracentrifugation. The viral titer was determined by infecting HEK293A cells with tenfold serially diluted virus and incubating them for 2 d. Immunofluorescence staining was then performed to count the infected cells and calculate the viral titer (infectious unit (i.u.)/mL)).

Western blot

The expression of the recombinant proteins was analyzed via Western blotting. Porcine kidney epithelial cells (PK-15, ATCC) were transduced with rAds at a multiplicity of infection (MOI) of 100. The cell lysates were harvested 48 h after transduction, and the protein samples were separated *via* 8% SDS–polyacrylamide gel electrophoresis (PAGE). For Western blot analysis, proteins in the gel were transferred to a nitrocellulose membrane. The membrane was blocked with 5% (w/*v*) skim milk in phosphate-buffered saline (PBS; Apsalagen, Bangkok, Thailand) containing 0.1% Tween-20 (PBST) for 1 h at room temperature. Primary antibody detection was performed using a mouse anti-V5 monoclonal antibody (Invitrogen) diluted in 2% skim milk/phosphate-buffered saline (PBS) for 3 h at room temperature. After being washed three times with PBST, the membrane was incubated with a horseradish peroxidase (HRP)-conjugated anti-mouse IgG secondary antibody (Abcam, Trumpington, Cambridge, UK) diluted in 2% skim milk/PBST for 90 min at room temperature. The membrane was washed three times before incubation with Westar SuperNova

chemiluminescent substrate (Cyanagen, Bologna, Italy) for 5 min. The signal was observed and imaged with a C-DiGit Blot Scanner (LI-COR Biosciences, Lincoln, NE, USA).

Immunofluorescence staining assay

HEK293A cells were infected with rAd at an MOI of 10, while PK-15 cells were transduced at an MOI of 100. The cells were fixed with 1% formaldehyde, incubated at room temperature for 10 min and then washed once with PBS prior to being permeabilized with 90% cold methanol for 5 min at 4°C. Following a PBS wash, the cells were blocked overnight at 4°C with PBS containing 2% (*v*/*v*) fetal bovine serum (FBS; Cytiva HyClone, Logan, UT, USA). Immunostaining was performed using a mouse anti-V5 monoclonal antibody (1:2,000; Invitrogen) or a goat anti-adenovirus polyclonal antibody (1:500; Abcam) diluted in PBS and incubated for 2 h at room temperature. After being washed twice with PBS, the cells were incubated with the following fluorophore-conjugated secondary antibodies for 1 h at room temperature: donkey anti-mouse IgG-Alexa Fluor® 594 (1:1,000; Abcam) or anti-goat IgG-NL493 (1:500; R&D Systems, Minneapolis, MN, USA). Following two washes with PBS, the nuclei were counterstained with 4′,6-diamidino-2-phenylindole (DAPI; Sigma–Aldrich, St. Louis, MO, USA) at a final concentration of 2 μM in PBS at room temperature for 5–10 min. Following a final wash with PBS, fluorescence images were acquired *via* an Olympus IX73 inverted fluorescence microscope (Olympus Life Science, Shinjuku City, Tokyo, Japan). All the imaging parameters were maintained constant across the experimental conditions to allow comparative analysis.

Animal ethics, animal use, and study location

The study was conducted on a large-scale commercial pig farm in Chai Nat Province, Thailand, with a history of PED-free herds. The study period is between October 2023 and February 2024, the winter and summer seasons, respectively. Five groups (n = 50) of crossbred sows (Large White × Landrace) with an average parity of 6 were randomly allocated after the same batch of artificial insemination and proof of pregnancy. There is no prescreening process for preexisting PEDV antibodies according to the history of a PED-free herd to minimize bias in randomized controlled trials. The groups were housed in an evaporative cooling house system in individual farrowing pens (2.5 × 2.5 m^2). A commercial diet (pregnant sow formula) was supplied at 2.5 kg per sow twice daily, with access to chlorinated water ad libitum. At the end of the study, the experimental sows and piglets were closely monitored for possible adverse reactions from the immunization. Furthermore, it was verified as disease-free and then returned to the farm production cycle.

All procedures involving sample collection were performed in accordance with the guidelines of the Institutional Animal Care and Use Committee (IACUC) of the Institute for Animals for Scientific Purpose Development and carried out under the regulation and permission of the Faculty of Veterinary Science, Mahidol University-Institute Animal Care and Use Committee (FVS-MU-IACUC-COA No. MUVS-2023–02-13), animal use license No. U1-01281–2558.

Preparation of sow colostrum/milk

The total sow colostrum/milk samples (n = 30) were randomly sampled from 5 sows in each group. The sow milk samples were collected during two different study periods: (i) the farrowing period and (ii) the postfarrowing period (5–9 days after birth). The concentrations of oxytocin used were 20 units/mL (X-TOCIN, BIC Chemical Co., Ltd., Nonthaburi, Thailand), 2 mL, and intramuscularly injected 10 min before the milk was collected. Aseptic techniques were applied at the udder area and the first two pairs of teats to reduce contamination. Then, the good-pulsation teats were gently squeezed to collect colostrum/milk samples in sterile containers. All colostrum/milk samples were stored at 4°C in the sterile repository and shipped immediately to the Virology and Cell Culture Unit, Faculty of Veterinary Science, Mahidol University. The colostrum/milk samples were transferred from sterile containers to 1.5 mL sterile microcentrifuge tubes, and high-speed centrifugation was performed for 10 min at 10,000 rpm at 4°C (MIKRO 200 R microliter centrifuge, Andreas Hettich GmbH & Co. KG, Tuttlingen, Germany). Centrifugation separates a colostrum/milk sample into three layers: a fatty layer (cream), a liquid phase (whey), and a pellet layer (protein and cells). The liquid phase was collected for further study.

Preparation of sow and piglet sera

The total piglet serum samples (n = 60) were randomly collected from two piglets from each of the five sows in each group. The piglets were gently restrained into a dorsal recumbency posture. Blood was subsequently collected from the anterior vena cava *via* a sterile 22G 1.5″ needle and 3 mL syringe (NIPRO (THAILAND) Co., Ltd., Phra Nakhon Si Ayutthaya, Thailand). Two milliliters of blood sample per piglet was stored in a blood collection serum tube (BD, Mississauga, Ontario, Canada). All blood samples were stored at 4°C in the sterile repository and shipped immediately to the Center of Veterinary Diagnosis, Faculty of Veterinary Science, Mahidol University. The blood samples were centrifuged for 10 min at 3,000 rpm (Universal Centrifuge PLC-036, Gemmy

Industrial Corp., Taipei, Taiwan). Sera were subsequently collected for further study.

Cells and PEDV preparation

PEDV was previously clinically isolated from PEDV-positive fecal samples by real-time polymerase chain reaction (RT–qPCR) (QuantStudio™ 3 Flex Real-Time PCR System (Thermo Fisher Scientific, USA)) of piglets displaying clinical signs in the Thailand outbreak in 2020. The sequence analysis revealed 100% identity with the complete genome sequence of the PEDV strain M3-SX2017, which has been deposited in the NCBI GenBank database under the accession number MK644603.1.

PEDV M3-SX2017 was propagated in Vero cells obtained from the American Type Culture Collection (ATCC: CCL-81). First, the Vero cells were cultured and maintained in cell growth medium prepared from a mixture of minimum essential medium (MEM; Gibco, Thermo Fisher Scientific, Grand Island, NE, USA) supplemented with 10% heat-inactivated fetal bovine serum (FBS; Gibco, Thermo Fisher Scientific, Grand Island, NE, USA), 2 mM L-glutamine, 200 units/mL penicillin G, 200 μg/mL streptomycin, and 0.25 μg/mL amphotericin B (Gibco, Thermo Fisher Scientific, Grand Island, NE, USA). Then, the Vero cells were cultured in a T25 tissue culture flask and washed twice with maintenance medium prepared from a mixture of MEM supplemented with 2% heat-inactivated FBS. A total of 200 μL of PEDV stock was inoculated in a T25 tissue culture flask and incubated under 5% CO_2 conditions for 2 h at 37 °C. Later, the virus inoculum was discarded, replaced with 5 mL maintenance medium, and incubated under 5% CO_2 conditions for 3–4 d at 37°C. The cytopathic effects (CPEs) in Vero cells demonstrated PEDV infection. The culture supernatant was used to quantify the infectious virus at the 50% tissue culture infective dose ($TCID_{50}$)/mL.

Neutralization assay

The serum and colostrum/milk samples were mixed thoroughly and aliquoted into microcentrifuge tubes. First, the samples were inactivated in a water bath operating for 30 min at 56°C. Then, 50 μL of diluent prepared from the mixture of MEM supplemented with 2% heat-inactivated FBS was loaded into a 96-well culture plate. The inactivated samples were loaded into the first column (1A-1H), and twofold serial dilutions were performed. The concentration of the PEDV M3-SX2017 stock was 10^3 $TCID_{50}$/50 μL, and the final concentration of the virus inoculum was 10^2 $TCID_{50}$/50 μL. Next, a total of 50 μL of virus inoculum was loaded into a 96-well culture plate and then incubated under 5% CO_2 and 60–80% humidity conditions for 90 min at 37°C. Vero cells were subcultured, counted, and diluted to a final concentration of 2×10^4 cells/50 μL. A total of 50 μL of Vero cell suspension was subsequently loaded into a 96-well culture plate and then incubated under 5% CO_2 and 60–80% humidity conditions for 3–5 d at 37°C. The highest concentration of serum and colostrum/milk samples that inhibited the CPEs of Vero cells was interpreted as the neutralization titer. To standardize the neutralization assay, the virus inoculum was subjected to a tenfold serial dilution as the back titration.

Detection of IgG and IgA using commercial PED ELISA kits

Commercial indirect ELISA kits were used to detect PEDV-specific IgG and IgA in the serum and colostrum/milk samples. The IgG level in the serum was examined via a PEDV IgG ELISA test kit (model EIA-SP002, Innoceleris LLC., Ames, IA, USA), whereas the IgG level in the colostrum/milk was examined via a PEDV IgG ELISA test kit (model EIA-MSG001, Innoceleris LLC., Ames, IA, USA). For IgA detection, the IgA level in the serum was evaluated via the PEDV IgA ELISA test kit (IDEXX, China), whereas the IgA level in the colostrum/milk samples was investigated via two different commercial PEDV IgA ELISA kits: EIA-MSA001 (Innoceleris LLC., Ames, IA, USA) and IDEXX (China). All the assays were conducted following the instructions provided by the manufacturers.

Detection of IgG and IgA *via* in-house ELISA

Recombinant COE and PEDVSME proteins were produced through transient gene expression in HEK293A cells via the polyethylenimine (PEI)-mediated transfection method. The transfection complex was prepared by combining the plasmid DNA encoding the COE or PEDVSME protein with PEI at an optimized ratio (1:1 DNA:PEI). The DNA–PEI complex was then introduced into cells at 70–80% confluency in DMEM supplemented with 10% FBS. Following 48–72 h of incubation with 5% CO_2 at 37°C, the secreted protein was harvested from the culture supernatant, clarified by centrifugation, and validated via Western blot analysis.

For ELISA, 96-well microplates (NEST, Wuxi, Jiangsu, China) were coated with 0.1 μg/well crude COE and PEDVSME proteins or 0.75 nmol of a single peptide in 50 μL of PBS. Following an overnight incubation at 4°C, the plates were washed twice with PBS containing 0.05% Tween 20 (PBST) (200 μL/well) and blocked with 5% FBS (Cytiva HyClone) in PBST (150 μL/well) for 1 h at room temperature. The samples were diluted in 1% FBS/PBST and incubated for 1 h at room temperature. Sample dilutions were optimized as follows: sow sera were diluted 1:100 and 1:20 for IgG and IgA detection, respectively, whereas piglet sera and colostrum samples were diluted

1:20 for both antibody isotypes. After three washes with PBST (200 μL/well), HRP-conjugated secondary antibodies were added to 1% FBS/PBST as follows: goat anti-pig IgG-HRP (1:20,000), goat anti-pig IgA-HRP (1:10,000), or a combination of both antibodies (1:10,000 final dilution) and incubated for 30 min at room temperature. After the samples were washed 3 times with PBST (200 μL/well), immunoreactivity was visualized via TMB substrate (70 μL/well; BioLegend, San Diego, CA, USA), with 15 min of incubation in the dark at room temperature. The reactions were stopped with 2 N sulfuric acid (H_2SO_4; 30 μL/well), and the absorbance was measured at 450 nm *via* a microplate reader (Thermo Scientific Multiskan FC, Thermo Fisher Scientific, Waltham, MA, USA).

Measurement of the cellular immune response *via* the ELISPOT assay

Antigen-specific IFN-γ-producing cells in sows before and after immunization were quantified via the Porcine IFN-γ Single-Color Enzyme ELISPOT Assay Kit (ImmunoSpot, Cleveland, OH, USA). Peripheral blood mononuclear cells (PBMCs) were isolated from sows and adjusted to 2×10^6 cells/mL. The cell suspension (100 μL) was aliquoted into each well of precoated IFN-γ plates previously washed once with 150 μL of PBS. The cells were cultured under two conditions: (i) restimulation with a peptide mixture (S1B-8, 15, 16.17, 19 and S2B-33 in 100 μL CTL-Test Medium) and (ii) without stimulation (no peptide added). One well of the plate was cultured with 2.5 μg of phytohemagglutinin (PHA-P; Sigma–Aldrich), which served as a positive control. The plates were then incubated undisturbed with 5% CO_2 for 24 h at 37°C. The staining step was then performed following the manufacturer's instructions. Spot-forming cells (SFCs) were enumerated via an ImmunoSpot EpiScan Reader and ImmunoSpot software (CTL). The results are reported as SFC per million cells after background values (negative control) were subtracted from the peptide-restimulated responses for each animal.

Statistical analysis

Statistical analyses were performed via GraphPad Prism V. 10.1.2 (GraphPad Software Inc., San Diego, CA, USA) and Jamovi V. 2.6.44 (The Jamovi Project, Sydney, Australia), which were retrieved from https://www.jamovi.org. An analysis of covariance (ANCOVA) was conducted. Prevaccination response levels were included as a covariate to control for baseline variations. Furthermore, a repeated measures ANCOVA was employed to evaluate the dynamics of antibody levels over time (at birth and weaning), with sow antibody titers serving as a covariate to assess the true efficacy of passive immunity transfer. A P value < 0.05 was considered statistically significant. For correlation analyses, the coefficient of determination (R^2) was calculated via standard curve interpolation in GraphPad Prism.

Abbreviations

Ab	Antibody
Ad	Adenovirus
Ad5	Human adenovirus type 5
CD4+	Cluster of differentiation 4/T helper cells
CD8+	Cluster of differentiation 8/Cytotoxic T cells
COE	Collagenase equivalent domain
CPEs	Cytopathic effects
CTB	Cholera toxin B subunit
DCs	Dendritic cells
DMEM	Dulbecco's modified Eagle's medium
DNA	Deoxyribonucleic acid
ELISA	Enzyme-linked immunosorbent assay
ELISpot	Enzyme-linked immunospot
FBS	Fetal bovine serum
GALTs	Gut-associated lymphoid tissues
GI	GM-CSF/IL-4 fusion protein
GM-CSF	Granulocyte–macrophage colony–stimulating factor
HEK-293	Human embryonic kidney 293
HRP	Horseradish peroxidase
IFN-γ	Interferon gamma
IgA	Immunoglobulin A
IgG	Immunoglobulin G
IL-4	Interleukin-4
i.u.	Infectious unit
LTB	Heat-labile enterotoxin B
MEM	Minimum essential medium
MOI	Multiplicity of infection
NT	Neutralizing antibody
OD450	Optical density at 450 nm
OmpH	Bacterial outer membrane protein H
ORFs	Open reading frames
PBS	Phosphate-buffered saline
PBMCs	Peripheral blood mononuclear cells
PBST	Phosphate-buffered saline containing Tween-20
PCV	Porcine circovirus
PEI	Polyethylenimine
PED	Porcine epidemic diarrhea
PEDV	Porcine epidemic diarrhea virus
PK-15	Porcine kidney epithelial cell 15
PRRSV	Porcine reproductive and respiratory syndrome virus
rAd	Recombinant adenovirus
rAd. PEDVSME	Recombinant adenovirus-expressing immunogen PEDVSME
RNA	Ribonucleic acid
sIgA	Secretory IgA
SFCs	Spot-forming cells
$TCID_{50}$	50% Tissue culture infectious dose
TMB	3,3′,5,5′-Tetramethylbenzidin
UTRs	Untranslated regions

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Authors' contributions

N.B. performed the experiments, analyzed and interpreted the data, wrote an original draft, and revised the manuscript. T.K. performed sample collection at a swine farm, conducted the experiments, analyzed and interpreted the data, wrote an original draft, and revised the manuscript. P.R. performed

the experiments and analyzed and interpreted the data. K.P. performed the experiments and analyzed and interpreted the data. P.C. and N.N. designed the experiments, provided resources, and acquired funding. Y.M.R. designed the experiments, analyzed and interpreted the data, provided resources and supervision, acquired funding, and reviewed and edited the manuscript. K.K. designed the clinical immunization study, conducted the experiments, provided resources and supervision, acquired funding, and reviewed and edited the manuscript. All the authors read the manuscript and provided comments on the final version of the manuscript.

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Data availability
Additional supporting data and materials are available from the corresponding author K.K. upon request.

Declarations

Ethics approval and consent to participate
The animal experiments were approved by the Veterinary Science, Mahidol University-Institute Animal Care and Use Committee (FVS-MU-IACUC-COA No. MUVS-2023–02-13), animal use license No. U1-01281–2558.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing interests.

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